

GLOBAL
EDITION



Ethics for the Information Age

SIXTH EDITION

Michael J. Quinn

ALWAYS LEARNING

PEARSON



6th edition

Global Edition

ETHICS

for the information age

This page is intentionally left blank.



6th edition
Global Edition

ETHICS

for the information age

MICHAEL J. QUINN
Seattle University

PEARSON

Boston Columbus Indianapolis New York San Francisco Upper Saddle River
Amsterdam Cape Town Dubai London Madrid Milan Munich Paris Montreal Toronto
Delhi Mexico City Sao Paulo Sydney Hong Kong Seoul Singapore Taipei Tokyo

Editorial Director, ECS Marcia Horton
Head of Learning Asset Acquisition, Global Edition Laura Dent
Acquisitions Editor Matt Goldstein
Program Manager Kayla Smith-Tarbox
Director of Marketing Christy Lesko
Marketing Manager Yezan Alayan
Marketing Assistant Jon Bryant
Director of Production Erin Gregg
Senior Managing Editor Scott Disanno
Acquisitions Editor, Global Edition Murchana Borthakur
Associate Project Editor, Global Edition Binita Roy
Project Manager Camille Trentacoste
Manufacturing Buyer Linda Sager
Senior Manufacturing Controller, Production, Global Edition Trudy Kimber
Text Designer Sandra Rigney
Manager, Text Permissions Tim Nicholls
Text Permission Project Manager William Opaluch
Cover Image © Tom Wang. Shutterstock
Media Project Manager Renata Butera
Full-Service Project Management Cypress Graphics, Paul C. Anagnostopoulos

Credits and acknowledgments borrowed from other sources and reproduced, with permission, appear on the appropriate page in the text.

Pearson Education Limited
Edinburgh Gate
Harlow
Essex CM20 2JE
England

and Associated Companies throughout the world

Visit us on the World Wide Web at: www.pearsonglobaleditions.com

© Pearson Education Limited 2015

The rights of Michael J. Quinn to be identified as the author of this work have been asserted by him in accordance with the Copyright, Designs and Patents Act 1988.

Authorized adaptation from the United States edition, entitled Ethics for the Information Age, 6th edition, ISBN 978-0-13-374162-9, by Michael J. Quinn, published by Pearson Education © 2015.

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without either the prior written permission of the publisher or a license permitting restricted copying in the United Kingdom issued by the Copyright Licensing Agency Ltd, Saffron House, 6–10 Kirby Street, London EC1N 8TS.

Many of the designations by manufacturers and sellers to distinguish their products are claimed as trademarks. Where those designations appear in this book, and the publisher was aware of a trademark claim, the designations have been printed in initial caps or all caps.

All trademarks used herein are the property of their respective owners. The use of any trademark in this text does not vest in the author or publisher any trademark ownership rights in such trademarks, nor does the use of such trademarks imply any affiliation with or endorsement of this book by such owners.

ISBN 10: 1-292-06123-5 ISBN 13: 978-1-292-06123-8 (Print) ISBN 13: 978-1-292-06190-0 (PDF)

British Library Cataloguing-in-Publication Data

A catalogue record for this book is available from the British Library

14 13 12 11 10 9 8 7 6 5 4 3 2 1

Typeset in Minion and Nofret by Cypress Graphics.

Printed and bound by Courier Westford in The United States of America.

Brief Contents

Preface 21

- 1 Catalysts for Change 29**
An Interview with Dalton Conley 75
- 2 Introduction to Ethics 77**
An Interview with James Moor 133
- 3 Networked Communications 137**
An Interview with Michael Liebhold 187
- 4 Intellectual Property 189**
An Interview with June Besek 251
- 5 Information Privacy 255**
An Interview with Michael Zimmer 293
- 6 Privacy and the Government 297**
An Interview with Jerry Berman 343
- 7 Computer and Network Security 347**
An Interview with Matt Bishop 385
- 8 Computer Reliability 389**
An Interview with Avi Rubin 433
- 9 Professional Ethics 435**
An Interview with Paul Axtell 475
- 10 Work and Wealth 479**
An Interview with Martin Ford 519
- Appendix A: Plagiarism 523**

Index 527

This page is intentionally left blank.

Contents

Preface 21

1 Catalysts for Change 29

1.1 Introduction 30

1.2 Milestones in Computing 33

1.2.1 Aids to Manual Calculating 33

1.2.2 Mechanical Calculators 34

1.2.3 Cash Register 36

1.2.4 Punched-Card Tabulation 37

1.2.5 Precursors of Commercial Computers 39

1.2.6 First Commercial Computers 41

1.2.7 Programming Languages and Time-Sharing 42

1.2.8 Transistor and Integrated Circuit 43

1.2.9 IBM System/360 45

1.2.10 Microprocessor 45

1.2.11 Personal Computer 46

1.3 Milestones in Networking 49

1.3.1 Electricity and Electromagnetism 49

1.3.2 Telegraph 51

1.3.3 Telephone 52

1.3.4 Typewriter and Teletype 53

1.3.5 Radio 53

1.3.6 Television 54

1.3.7 Remote Computing 56

1.3.8 ARPANET 56

1.3.9 Email 57

1.3.10 Internet 57

1.3.11 NSFNET 59

1.3.12 Broadband 59

1.3.13 Wireless Networks 59

1.4 Milestones in Information Storage and Retrieval 60

- 1.4.1 Greek Alphabet 60
- 1.4.2 Codex and Paper 60
- 1.4.3 Gutenberg's Printing Press 61
- 1.4.4 Newspapers 61
- 1.4.5 Hypertext 61
- 1.4.6 Graphical User Interface 62
- 1.4.7 Single-Computer Hypertext Systems 64
- 1.4.8 Networked Hypertext: World Wide Web 64
- 1.4.9 Search Engines 65

1.5 Information Technology Issues 66

- Summary 67
- Review Questions 69
- Discussion Questions 70
- In-Class Exercises 71
- Further Reading and Viewing 72
- References 72

An Interview with Dalton Conley 75**2 Introduction to Ethics 77****2.1 Introduction 77**

- 2.1.1 Defining Terms 78
- 2.1.2 Four Scenarios 80
- 2.1.3 Overview of Ethical Theories 83

2.2 Subjective Relativism 84

- 2.2.1 The Case for Subjective Relativism 84
- 2.2.2 The Case against Subjective Relativism 85

2.3 Cultural Relativism 86

- 2.3.1 The Case for Cultural Relativism 87
- 2.3.2 The Case against Cultural Relativism 88

2.4 Divine Command Theory 90

- 2.4.1 The Case for the Divine Command Theory 90
- 2.4.2 The Case against the Divine Command Theory 91

2.5 Ethical Egoism 93

- 2.5.1 The Case for Ethical Egoism 93
- 2.5.2 The Case against Ethical Egoism 94

2.6 Kantianism 95

- 2.6.1 Good Will and the Categorical Imperative 96
- 2.6.2 Evaluating a Scenario Using Kantianism 98

2.6.3	The Case for Kantianism	99
2.6.4	The Case against Kantianism	99
2.7	Act Utilitarianism	100
2.7.1	Principle of Utility	101
2.7.2	Evaluating a Scenario Using Act Utilitarianism	102
2.7.3	The Case for Act Utilitarianism	103
2.7.4	The Case against Act Utilitarianism	104
2.8	Rule Utilitarianism	106
2.8.1	Basis of Rule Utilitarianism	106
2.8.2	Evaluating a Scenario Using Rule Utilitarianism	106
2.8.3	The Case for Rule Utilitarianism	107
2.8.4	The Case against Utilitarianism in General	108
2.9	Social Contract Theory	109
2.9.1	The Social Contract	109
2.9.2	Rawls's Theory of Justice	111
2.9.3	Evaluating a Scenario Using Social Contract Theory	113
2.9.4	The Case for Social Contract Theory	114
2.9.5	The Case against Social Contract Theory	115
2.10	Virtue Ethics	117
2.10.1	Virtues and Vices	117
2.10.2	Making a Decision Using Virtue Ethics	119
2.10.3	The Case for Virtue Ethics	120
2.10.4	The Case against Virtue Ethics	121
2.11	Comparing Workable Ethical Theories	122
2.12	Morality of Breaking the Law	124
2.12.1	Social Contract Theory Perspective	124
2.12.2	Kantian Perspective	124
2.12.3	Rule Utilitarian Perspective	125
2.12.4	Act Utilitarian Perspective	125
2.12.5	Conclusion	126
	Summary	126
	Review Questions	127
	Discussion Questions	129
	In-Class Exercises	130
	Further Reading and Viewing	131
	References	131
	An Interview with James Moor	133

3 Networked Communications 137

- 3.1 Introduction 137**
- 3.2 Spam 139**
 - 3.2.1 The Spam Epidemic 140
 - 3.2.2 Need for Social-Technical Solutions 141
 - 3.2.3 Case Study: Ann the Acme Accountant 142
- 3.3 Internet Interactions 145**
 - 3.3.1 The World Wide Web 145
 - 3.3.2 The Rise of the App 145
 - 3.3.3 How We Use the Internet 145
- 3.4 Text Messaging 148**
 - 3.4.1 Transforming Lives in Developing Countries 148
 - 3.4.2 Twitter 149
 - 3.4.3 Business Promotion 149
 - 3.4.4 Political Activism 149
- 3.5 Censorship 150**
 - 3.5.1 Direct Censorship 150
 - 3.5.2 Self-Censorship 151
 - 3.5.3 Challenges Posed by the Internet 152
 - 3.5.4 Government Filtering and Surveillance of Internet Content 152
 - 3.5.5 Ethical Perspectives on Censorship 153
- 3.6 Freedom of Expression 155**
 - 3.6.1 History 155
 - 3.6.2 Freedom of Expression Not an Absolute Right 156
 - 3.6.3 *FCC v. Pacifica Foundation* 157
 - 3.6.4 Case Study: Kate's Blog 158
- 3.7 Children and Inappropriate Content 160**
 - 3.7.1 Web Filters 160
 - 3.7.2 Child Internet Protection Act 161
 - 3.7.3 Ethical Evaluations of CIPA 162
 - 3.7.4 Sexting 163
- 3.8 Breaking Trust 165**
 - 3.8.1 Identity Theft 165
 - 3.8.2 Chat-Room Predators 166
 - 3.8.3 Ethical Evaluations of Police Sting Operations 167
 - 3.8.4 False Information 169
 - 3.8.5 Cyberbullying 170
- 3.9 Internet Addiction 171**
 - 3.9.1 Is Internet Addiction Real? 171
 - 3.9.2 Contributing Factors 173

3.9.3	Ethical Evaluation of Internet Addiction	174
	Summary	175
	Review Questions	176
	Discussion Questions	177
	In-Class Exercises	179
	Further Reading and Viewing	180
	References	181

	An Interview with Michael Liebhold	187
--	---	-----

4 Intellectual Property 189

4.1	Introduction	189
4.2	Intellectual Property Rights	191
4.2.1	Property Rights	191
4.2.2	Extending the Argument to Intellectual Property	192
4.2.3	Benefits of Intellectual Property Protection	195
4.2.4	Limits to Intellectual Property Protection	195
4.3	Protecting Intellectual Property	197
4.3.1	Trade Secrets	197
4.3.2	Trademarks and Service Marks	198
4.3.3	Patents	198
4.3.4	Copyrights	200
4.4	Fair Use	204
4.4.1	<i>Sony v. Universal City Studios</i>	205
4.4.2	Digital Recording Technology	207
4.4.3	Audio Home Recording Act of 1992	207
4.4.4	<i>RIAA v. Diamond Multimedia</i>	207
4.4.5	<i>Kelly v. Arriba Soft</i>	208
4.4.6	Google Books	209
4.5	New Restrictions on Use	210
4.5.1	Digital Millennium Copyright Act	211
4.5.2	Digital Rights Management	212
4.5.3	Secure Digital Music Initiative	212
4.5.4	Sony BMG Music Entertainment Rootkit	212
4.5.5	Encrypting DVDs	213
4.5.6	Foiling HD-DVD Encryption	214
4.5.7	Criticisms of Digital Rights Management	214
4.5.8	Online Music Stores Drop Digital Rights Management	215
4.5.9	Microsoft Xbox One	215
4.6	Peer-to-Peer Networks and Cyberlockers	216
4.6.1	Napster	217

4.6.2	FastTrack	217
4.6.3	BitTorrent	218
4.6.4	RIAA Lawsuits	219
4.6.5	<i>MGM v. Grokster</i>	220
4.6.6	Legal Action against the Pirate Bay	222
4.6.7	PRO-IP Act	223
4.6.8	Megaupload Shutdown	223
4.6.9	Legal Music Services on the Internet	223
4.7	Protections for Software	224
4.7.1	Software Copyrights	224
4.7.2	Violations of Software Copyrights	224
4.7.3	Safe Software Development	225
4.7.4	Software Patents	226
4.8	Open-Source Software	228
4.8.1	Consequences of Proprietary Software	228
4.8.2	“Open Source” Definition	229
4.8.3	Beneficial Consequences of Open-Source Software	230
4.8.4	Examples of Open-Source Software	230
4.8.5	The GNU Project and Linux	231
4.8.6	Impact of Open-Source Software	232
4.8.7	Critique of the Open-Source Software Movement	232
4.9	Legitimacy of Intellectual Property Protection for Software	233
4.9.1	Rights-Based Analysis	233
4.9.2	Utilitarian Analysis	234
4.9.3	Conclusion	235
4.10	Creative Commons	236
	Summary	238
	Review Questions	241
	Discussion Questions	242
	In-Class Exercises	242
	Further Reading and Viewing	243
	References	244
	An Interview with June Besek	251

5 Information Privacy 255

5.1	Introduction	255
5.2	Perspectives on Privacy	257
5.2.1	Defining Privacy	257
5.2.2	Harms and Benefits of Privacy	258
5.2.3	Is There a Natural Right to Privacy?	260

5.2.4	Privacy and Trust	264
5.2.5	Case Study: The New Parents	264
5.3	Information Disclosures	266
5.3.1	Facebook Tags	268
5.3.2	Enhanced 911 Services	268
5.3.3	Rewards or Loyalty Programs	269
5.3.4	Body Scanners	269
5.3.5	RFID Tags	270
5.3.6	Implanted Chips	271
5.3.7	OnStar	272
5.3.8	Automobile “Black Boxes”	272
5.3.9	Medical Records	273
5.3.10	Digital Video Recorders	273
5.3.11	Cookies and Flash Cookies	273
5.4	Data Mining	274
5.4.1	Data Mining Defined	274
5.4.2	Opt-In versus Opt-Out Policies	275
5.4.3	Examples of Data Mining	276
5.4.4	Social Network Analysis	279
5.5	Examples of Consumer Backlash	280
5.5.1	Marketplace: Households	280
5.5.2	Facebook Beacon	281
5.5.3	Netflix Prize	281
5.5.4	Malls Track Shoppers’ Cell Phones	282
5.5.5	iPhone Apps Uploading Address Books	282
5.5.6	Instagram’s Proposed Change to Terms of Service	283
	Summary	283
	Review Questions	284
	Discussion Questions	285
	In-Class Exercises	287
	Further Reading and Viewing	288
	References	289
	An Interview with Michael Zimmer	293

6 Privacy and the Government 297

6.1	Introduction	297
6.2	US Legislation Restricting Information Collection	299
6.2.1	Employee Polygraph Protection Act	299
6.2.2	Children’s Online Privacy Protection Act	299
6.2.3	Genetic Information Nondiscrimination Act	299

6.3	Information Collection by the Government	300
6.3.1	Census Records	300
6.3.2	Internal Revenue Service Records	301
6.3.3	FBI National Crime Information Center 2000	302
6.3.4	OneDOJ Database	303
6.3.5	Closed-Circuit Television Cameras	304
6.3.6	Police Drones	305
6.4	Covert Government Surveillance	306
6.4.1	Wiretaps and Bugs	306
6.4.2	Operation Shamrock	309
6.4.3	Carnivore Surveillance System	310
6.4.4	Covert Activities after 9/11	310
6.5	US Legislation Authorizing Wiretapping	311
6.5.1	Title III	311
6.5.2	Foreign Intelligence Surveillance Act	312
6.5.3	Electronic Communications Privacy Act	312
6.5.4	Stored Communications Act	313
6.5.5	Communications Assistance for Law Enforcement Act	313
6.6	USA PATRIOT Act	314
6.6.1	Provisions of the Patriot Act	314
6.6.2	National Security Letters	315
6.6.3	Responses to the Patriot Act	316
6.6.4	Successes and Failures	317
6.6.5	Patriot Act Renewal	318
6.6.6	Long-Standing NSA Access to Telephone Records	318
6.7	Regulation of Public and Private Databases	319
6.7.1	Code of Fair Information Practices	319
6.7.2	Privacy Act of 1974	321
6.7.3	Fair Credit Reporting Act	322
6.7.4	Fair and Accurate Credit Transactions Act	322
6.7.5	Financial Services Modernization Act	322
6.8	Data Mining by the Government	323
6.8.1	Internal Revenue Service Audits	323
6.8.2	Syndromic Surveillance Systems	323
6.8.3	Telecommunications Records Database	323
6.8.4	Predictive Policing	324
6.9	National Identification Card	324
6.9.1	History and Role of the Social Security Number	324
6.9.2	Debate over a National ID Card	325
6.9.3	The REAL ID Act	327

6.10	Information Dissemination	328
6.10.1	Family Education Rights and Privacy Act	328
6.10.2	Video Privacy Protection Act	328
6.10.3	Health Insurance Portability and Accountability Act	329
6.10.4	Freedom of Information Act	329
6.10.5	Toll Booth Records Used in Court	330
6.11	Invasion	330
6.11.1	Telemarketing	331
6.11.2	Loud Television Commercials	331
6.11.3	Requiring Identification for Pseudoephedrine Purchases	331
6.11.4	Advanced Imaging Technology Scanners	332
	Summary	333
	Review Questions	334
	Discussion Questions	336
	In-Class Exercises	337
	Further Reading and Viewing	337
	References	337
	An Interview with Jerry Berman	343

7 Computer and Network Security 347

7.1	Introduction	347
7.2	Hacking	348
7.2.1	Hackers, Past and Present	348
7.2.2	Penalties for Hacking	349
7.2.3	Selected Hacking Incidents	350
7.2.4	Case Study: Firesheep	351
7.3	Malware	353
7.3.1	Viruses	354
7.3.2	The Internet Worm	356
7.3.3	Sasser	360
7.3.4	Instant Messaging Worms	361
7.3.5	Conficker	361
7.3.6	Cross-Site Scripting	361
7.3.7	Drive-By Downloads	361
7.3.8	Trojan Horses and Backdoor Trojans	362
7.3.9	Rootkits	362
7.3.10	Spyware and Adware	362
7.3.11	Bots and Botnets	362
7.3.12	Defensive Measures	363

7.4	Cyber Crime and Cyber Attacks	363
7.4.1	Phishing and Spear Phishing	364
7.4.2	SQL Injection	364
7.4.3	Denial-of-Service and Distributed Denial-of-Service Attacks	364
7.4.4	Cyber Crime	365
7.4.5	Politically Motivated Cyber Attacks	367
7.5	Online Voting	371
7.5.1	Motivation for Online Voting	371
7.5.2	Proposals	372
7.5.3	Ethical Evaluation	373
	Summary	376
	Review Questions	376
	Discussion Questions	377
	In-Class Exercises	379
	Further Reading and Viewing	380
	References	380

	An Interview with Matt Bishop	385
--	--------------------------------------	------------

8 Computer Reliability 389

8.1	Introduction	389
8.2	Data Entry or Data Retrieval Errors	390
8.2.1	Disenfranchised Voters	390
8.2.2	False Arrests	390
8.2.3	Utilitarian Analysis: Accuracy of NCIC Records	391
8.3	Software and Billing Errors	392
8.3.1	Errors Leading to System Malfunctions	392
8.3.2	Errors Leading to System Failures	393
8.3.3	Analysis: E-retailer Posts Wrong Price, Refuses to Deliver	395
8.4	Notable Software System Failures	395
8.4.1	Patriot Missile	396
8.4.2	Ariane 5	397
8.4.3	AT&T Long-Distance Network	398
8.4.4	Robot Missions to Mars	399
8.4.5	Denver International Airport	400
8.4.6	Tokyo Stock Exchange	401
8.4.7	Direct Recording Electronic Voting Machines	402
8.5	Therac-25	405
8.5.1	Genesis of the Therac-25	405
8.5.2	Chronology of Accidents and AECL Responses	406
8.5.3	Software Errors	408
8.5.4	Postmortem	410

8.5.5	Moral Responsibility of the Therac-25 Team	411
8.5.6	Postscript	412
8.6	Computer Simulations	412
8.6.1	Uses of Simulation	412
8.6.2	Validating Simulations	414
8.7	Software Engineering	415
8.7.1	Specification	416
8.7.2	Development	416
8.7.3	Validation	417
8.7.4	Evolution	418
8.7.5	Software Quality Is Improving	418
8.8	Software Warranties and Vendor Liability	419
8.8.1	Shrink-wrap Warranties	419
8.8.2	Are Software Warranties Enforceable?	420
8.8.3	Should Software Be Considered a Product?	423
8.8.4	Case Study: Incredible Bulk	423
	Summary	424
	Review Questions	426
	Discussion Questions	427
	In-Class Exercises	428
	Further Reading and Viewing	429
	References	429
	An Interview with Avi Rubin	433

9 Professional Ethics 435

9.1	Introduction	435
9.2	How Well Developed Are the Computing Professions?	437
9.2.1	Characteristics of a Fully Developed Profession	437
9.2.2	Case Study: Certified Public Accountants	438
9.2.3	How Do Computer-Related Careers Stack Up?	439
9.3	Software Engineering Code of Ethics	441
9.4	Analysis of the Code	449
9.4.1	Preamble	449
9.4.2	Alternative List of Fundamental Principles	450
9.5	Case Studies	451
9.5.1	Software Recommendation	452
9.5.2	Child Pornography	453
9.5.3	Antiworm	454
9.5.4	Consulting Opportunity	456

9.6 Whistle-Blowing	458
9.6.1 Morton Thiokol/NASA	459
9.6.2 Hughes Aircraft	460
9.6.3 US Legislation Related to Whistle-Blowing	462
9.6.4 Morality of Whistle-Blowing	463
Summary	466
Review Questions	467
Discussion Questions	468
In-Class Exercises	470
Further Reading and Viewing	471
References	471

An Interview with Paul Axtell	475
--------------------------------------	-----

10 Work and Wealth 479

10.1 Introduction	479
10.2 Automation and Employment	480
10.2.1 Automation and Job Destruction	481
10.2.2 Automation and Job Creation	483
10.2.3 Effects of Increase in Productivity	484
10.2.4 Rise of the Robots?	486
10.3 Workplace Changes	489
10.3.1 Organizational Changes	490
10.3.2 Telework	491
10.3.3 Temporary Work	493
10.3.4 Monitoring	493
10.3.5 Multinational Teams	494
10.4 Globalization	496
10.4.1 Arguments for Globalization	496
10.4.2 Arguments against Globalization	497
10.4.3 Dot-Com Bust Increases IT Sector Unemployment	498
10.4.4 Foreign Workers in the American IT Industry	498
10.4.5 Foreign Competition	499
10.5 The Digital Divide	500
10.5.1 Global Divide	501
10.5.2 Social Divide	502
10.5.3 Models of Technological Diffusion	502
10.5.4 Critiques of the Digital Divide	504
10.5.5 Massive Open Online Courses	505
10.5.6 Net Neutrality	506

10.6 The “Winner-Take-All Society”	507
10.6.1 Harmful Effects of Winner-Take-All	508
10.6.2 Reducing Winner-Take-All Effects	510
Summary	510
Review Questions	512
Discussion Questions	512
In-Class Exercises	514
Further Reading and Viewing	515
References	515

An Interview with Martin Ford	519
--------------------------------------	------------

Appendix A: Plagiarism 523

Consequences of Plagiarism	523
Types of Plagiarism	523
Guidelines for Citing Sources	524
How to Avoid Plagiarism	524
Misuse of Sources	524
Additional Information	525
References	525

Index	527
--------------	------------

This page is intentionally left blank.

Preface

Computers and high-speed communication networks are transforming our world. These technologies have brought us many benefits, but they have also raised many social and ethical concerns. My view is that we ought to approach every new technology in a thoughtful manner, considering not just its short-term benefits, but also how its long-term use will affect our lives. A thoughtful response to information technology requires a basic understanding of its history, an awareness of current information-technology-related issues, and a familiarity with ethics. I have written *Ethics for the Information Age* with these ends in mind.

Ethics for the Information Age is suitable for college students at all levels. The only prerequisite is some experience using computers and the Internet. The book is appropriate for a stand-alone “computers and society” or “computer ethics” course offered by a computer science, business, or philosophy department. It can also be used as a supplemental textbook in a technical course that devotes some time to social and ethical issues related to computing.

As students discuss controversial issues related to information technology, they have the opportunity to learn from one other and improve their critical thinking skills. The provocative questions raised at the end of every chapter, together with dozens of in-class exercises, provide many opportunities for students to express their viewpoints. My hope is that they will get better at evaluating complex issues and defending their conclusions with facts, sound values, and rational arguments.

WHAT’S NEW IN THE SIXTH EDITION

The most significant change in the sixth edition is the new emphasis on virtue ethics. I have written a completely new section on virtue ethics that appears in Chapter 2, replacing the description of virtue ethics that formerly appeared in the chapter on professional ethics. In addition, I have included analyses from the perspective of virtue ethics to the case studies appearing in Chapters 3, 5, and 7.

To increase the relevance and value of the “Further Reading and Viewing” sections, I have eliminated the references to scholarly tomes. They have been replaced by lists of recent magazine and newspaper articles, television interviews, documentaries, and other videos available on the Internet. Most of the videos are only a few minutes long and can fuel interesting classroom discussions.

In response to a suggestion from one of the reviewers, I have added a table to Chapter 7 that provides students with practical tips about how to choose good passwords.

The sixth edition references many important recent developments; among them are:

- Edward Snowden’s revelations of longstanding National Security Agency access to telephone metadata, email messages, and live communications;
- the privacy implications of Twitter, Foursquare, Instagram, and other apps gathering information from address books stored on smartphones;
- the controversy surrounding Microsoft’s proposal for digital rights management on the Xbox One;
- the activities of the “hacktivist” group Anonymous;
- benefits and harms of tracking the movement of people through their smartphones;
- the debate over the use of drones by police departments;
- retailers using information collected from online sales to differentiate between customers and offer different prices to different people;
- retailers using targeted direct marketing to win new customers;
- the use of “crowdsourcing” by companies to improve products and services;
- coverage of how cell phones are changing lives in developing countries;
- predictive policing based on data mining;
- massive open online courses (MOOCs) and implications for students from different socio-economic groups; and
- the “Internet of Things”—Internet-connected devices that can be controlled remotely.

Finally, I have updated facts and figures throughout the book.

ORGANIZATION OF THE BOOK

The book is divided into ten chapters. Chapter 1 has three objectives: to get the reader thinking about the process of technological change; to present a brief history of computing, networking, and information storage and retrieval; and to provide examples of moral problems brought about by the introduction of information technology.

Chapter 2 is an introduction to ethics. It presents nine different theories of ethical decision-making, weighing the pros and cons of each one. Five of these theories—Kantianism, act utilitarianism, rule utilitarianism, social contract theory, and virtue ethics—are deemed the most appropriate “tools” for analyzing moral problems in the remaining chapters.

Chapters 3–10 discuss a wide variety of issues related to the introduction of information technology into society. I think of these chapters as forming concentric rings around a particular computer user.

Chapter 3 is the innermost ring, dealing with what can happen when people communicate over the Internet using the Web, email, and Twitter. Issues such as the increase in spam, easy access to pornography, cyberbullying, and Internet addiction raise important questions related to quality of life, free speech, and censorship.

The next ring, Chapter 4, deals with the creation and exchange of intellectual property. It discusses intellectual property rights, legal safeguards for intellectual property, the definition of fair use, digital rights management, abuses of peer-to-peer networks, the rise of the open-source movement, and the legitimacy of intellectual property protection for software.

Chapter 5 focuses on information privacy. What is privacy exactly? Is there a natural right to privacy? How do others learn so much about us? The chapter describes the electronic trail that people leave behind when they use a cell phone, make credit card purchases, open a bank account, go to a physician, or apply for a loan.

Chapter 6 focuses on privacy and the US government. Using Daniel Solove's taxonomy of privacy as our organizing principle, we look at how the government has steered between the competing interests of personal privacy and public safety. We consider US legislation to restrict information collection and government surveillance; government regulation of private databases and abuses of large government databases; legislation to reduce the dissemination of information and legislation that has had the opposite effect; and finally government actions to prevent the invasion of privacy as well as invasive government actions. Along the way, we discuss the implications of the USA PATRIOT Act and the debate over the REAL ID Act to establish a de facto national identification card.

Chapter 7 focuses on the vulnerabilities of networked computers. A case study focuses on the release of the Firesheep extension to the Firefox Web browser. A section on malware discusses rootkits, spyware, cross-site scripting, and drive-by downloads. We discuss common Internet-based attacks—phishing, spear-phishing, SQL injection, denial-of-service attacks, and distributed denial-of-service attacks—and how they are used for cyber crime, cyber espionage, and cyber attacks. We conclude with a discussion of the risks associated with online voting.

Computerized system failures have led to lost business, the destruction of property, human suffering, and even death. Chapter 8 describes some notable software system failures, including the story of the Therac-25 radiation therapy system. It also discusses the reliability of computer simulations, the emergence of software engineering as a distinct discipline, and the validity of software warranties.

Chapter 9 is particularly relevant for those readers who plan to take jobs in the computer industry. The chapter presents a professional code related to computing, the Software Engineering Code of Ethics and Professional Practice, followed by an analysis of the code. Several case studies illustrate how to use the Software Engineering Code of Ethics and Professional Practice to evaluate moral problems related to the use of computers. The chapter concludes with an ethical evaluation of whistle-blowing, an extreme example of organizational dissent.

Chapter 10 raises a wide variety of issues related to how information technology has impacted work and wealth. Topics include workplace monitoring, telecommuting, and globalization. Does automation increase unemployment? Is there a “digital divide” separating society into “haves” and “have nots”? Is information technology widening the gap between rich and poor? These are just a few of the important questions the chapter addresses.

<i>Unit</i>	<i>Name</i>	<i>Chapter(s)</i>
SP1	History of computing	1
SP2	Social context of computing	1, 3, 10
SP3	Methods and tools of analysis	2–10
SP4	Professional and ethical responsibilities	9
SP5	Risks and liabilities of computer-based systems	8
SP6	Intellectual property	4
SP7	Privacy and civil liberties	5, 6
SP8	Computer crime	3, 7
SP9	Economic issues in computing	10
SP10	Philosophical frameworks	2

TABLE 1 Mapping between the units of the Social and Professional Issues course in *Computing Curricula 2001* and the chapters of this book.

NOTE TO INSTRUCTORS

In December 2001, a joint task force of the IEEE Computer Society and the Association for Computing Machinery released the final draft of *Computing Curricula 2001* (www.computer.org/education/cc2001/final). The report recommends that every undergraduate computer science degree program incorporate 40 hours of instruction related to social and professional issues related to computing. For those departments that choose to dedicate an entire course to these issues, the report provides a model syllabus for CS 280T, Social and Professional Issues. *Ethics for the Information Age* covers all of the major topics listed in the syllabus. Table 1 shows the mapping between the 10 units of CS 280T and the chapters of this book.

The organization of the book makes it easy to adapt to your particular needs. If your syllabus does not include the history of information technology, you can skip the middle three sections of Chapter 1 and still expose your students to examples motivating the formal study of ethics in Chapter 2. After Chapter 2, you may cover the remaining chapters in any order you choose, because Chapters 3–10 do not depend on one other.

Many departments choose to incorporate discussions of social and ethical issues throughout the undergraduate curriculum. The independence of Chapters 3–10 makes it convenient to use *Ethics for the Information Age* as a supplementary textbook. You can simply assign readings from the chapters most closely related to the course topic.

SUPPLEMENTS

The following supplements are available to qualified instructors on Pearson's Instructor Resource Center. Please contact your local Pearson sales representative or visit www.pearsonglobaleditions.com/Quinn to access this material.

- An instructor's manual provides tips for teaching a course in computer ethics. It also contains answers to all of the review questions.

- A test bank contains more than 300 multiple-choice, fill-in-the-blank, and essay questions that you can use for quizzes, midterms, and final examinations.
- A set of PowerPoint lecture slides outlines the material covered in every chapter.

FEEDBACK

Ethics for the Information Age cites hundreds of sources and includes dozens of ethical analyses. Despite my best efforts and those of many reviewers, the book is bound to contain errors. I appreciate getting comments (both positive and negative), corrections, and suggestions from readers. Please send them to quinnm@seattleu.edu or Michael J. Quinn, Seattle University, College of Science and Engineering, 901 12th Avenue, Seattle, WA 98122.

ACKNOWLEDGMENTS

I appreciate the continuing support of a great publications team, beginning with editor Matt Goldstein, and including Kayla Tarbox, Marilyn Lloyd, Kathy Ringrose, Craig Jones, Paul Anagnostopoulos, Jacqui Scarlott, Jennifer McClain, Richard Camp, Laurel Muller, and Ted Laux. I thank them and everyone else who helped produce this edition.

A superb group of reviewers provided me with many helpful suggestions regarding new material to incorporate into the sixth edition. I thank: Donna Maria D'Ambrosio, University of South Florida; Dawit Demissie, The Sage Colleges; J.C. Diaz, The University of Tulsa; Fred Geldon, George Mason University; Richard E. Gordon, University of Delaware; Michael Gourley, University of Central Oklahoma; Robert Greene, University of Wisconsin–Eau Claire; Gurdeep Hura, University of Maryland Eastern Shore; Evelyn Lulis, DePaul University; Aparna Mahadev, Worcester State University; Daniel Palmer, Kent State University; Jason Rogers, George Mason University; and Scott Vitz, Indiana University–Purdue University Fort Wayne.

I am grateful to philosophy professor Maria Carl of Seattle University, who reviewed the new material on virtue ethics and provided me with valuable and constructive feedback.

I thank all who participated in the creation of the first five editions or provided useful suggestions for the sixth edition: Paul Anagnostopoulos, Valerie Anctil, Beth Anderson, Bob Baddeley, George Beekman, Brian Breck, Sherry Clark, Thomas Dietrich, Roger Eastman, Beverly Fusfield, Robert Greene, Peter Harris, Susan Hartman, Michael Hirsch, Michael Johnson, Marilyn Lloyd, Pat McCutcheon, Beth Paquin, Brandon Quinn, Stuart Quinn, Victoria Quinn, Charley Renn, Lindsey Triebel, and Shauna Weaver.

Reviewers of previous editions include: Ramprasad Bala, University of Massachusetts at Dartmouth; Phillip Barry, University of Minnesota; Bo Brinkman, Miami University; Diane Cassidy, The University of North Carolina at Charlotte; Madhavi M. Chakrabarty, New Jersey Institute of Technology; John Clark, University of Colorado at Denver; Timothy Colburn, University of Minnesota–Duluth; Lorrie Faith Cranor, Carnegie Mellon University; Lee D. Cornell, Minnesota State University, Mankato; Richard W. Egan, New Jersey Institute of Technology; David Goodall, State University of New York at Albany; Richard E. Gordon, University of Delaware; Mike Gourley,

University of Central Oklahoma; Fritz H. Grupe, University of Nevada, Reno; Ric Heishman, George Mason University; Paulette Kidder, Seattle University; Evelyn Lulis, DePaul University; Tamara A. Maddox, George Mason University; Eric Manley, Drake University; Richard D. Manning, Nova Southeastern University; John G. Messerly, University of Texas at Austin; Joe Oldham, Centre College; Mimi Opkins, California State University, Long Beach; Holly Patterson-McNeill, Lewis-Clark State College; Colin Potts, Georgia Tech; Medha S. Sarkar, Middle Tennessee State University; Michael Scanlan, Oregon State University; Robert Sloan, University of Illinois at Chicago; Matthew Stockton, Portland Community College; Leon Tabak, Cornell College; Renée Turban, Arizona State University; Scott Vitz, Indiana University–Purdue University Fort Wayne; David Womack, University of Texas at San Antonio; John Wright, Juniata College; and Matthew Zullo, Wake Technical Community College.

Finally, I am indebted to my wife Victoria for her support and encouragement. You are a wonderful helpmate. Thanks for everything.

Michael J. Quinn
Seattle, Washington

Pearson wishes to thank and acknowledge the following people for their work on the Global Edition:

Contributor

Sapna Sinha, Amity Institute of Information Technology

Reviewers

B. R. Chandavarkar, National Institute of Technology Karnataka, Surathkal
Mohit P. Tahlani, National Institute of Technology Karnataka, Surathkal
Sahil Raj, Punjabi University, Patiala

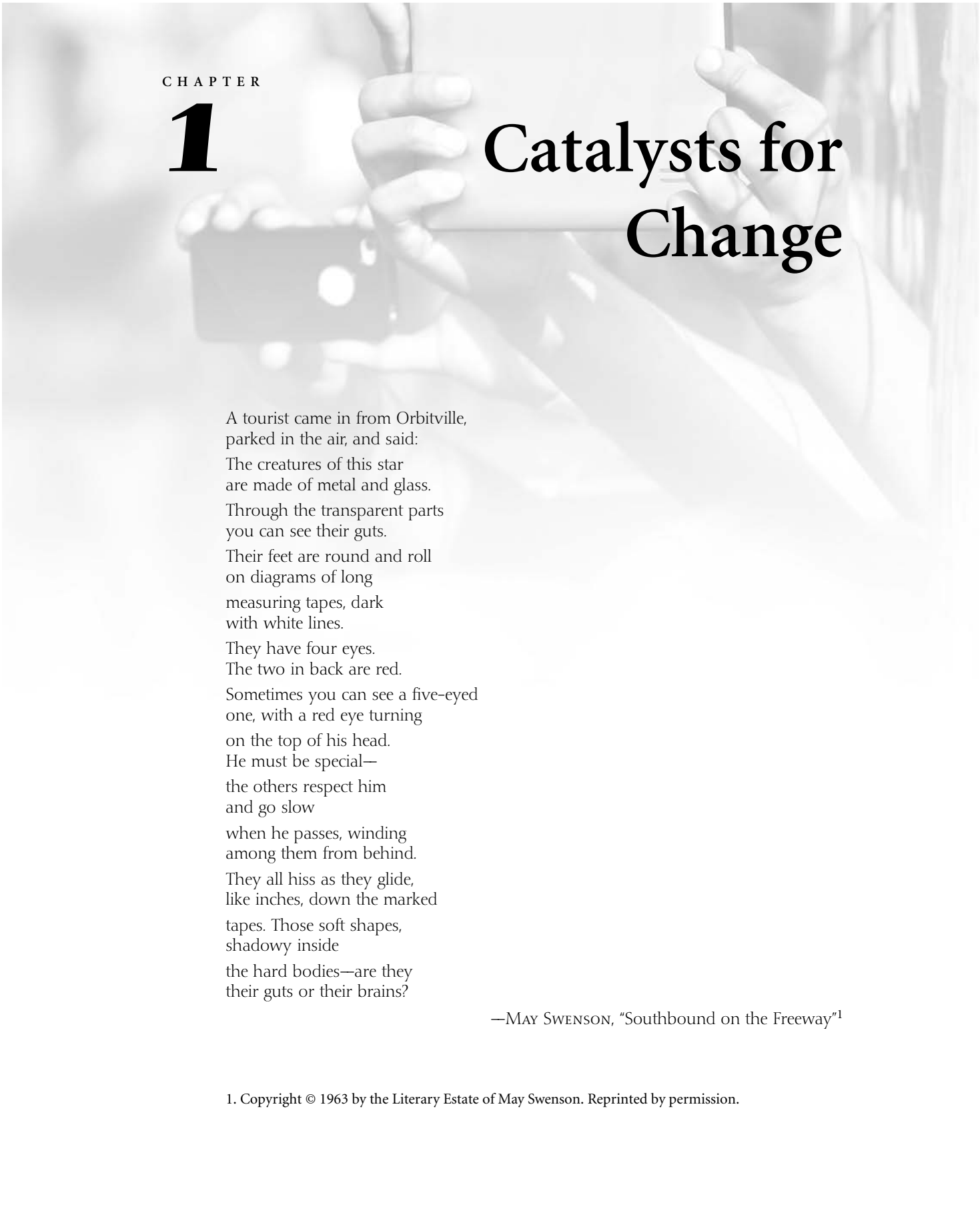
We never know how high we are
Till we are called to rise;
And then, if we are true to plan,
Our staturs touch the skies.
The heroism we recite
Would be a daily thing
Did not ourselves the cubits warp
For fear to be a king.

—EMILY DICKINSON, *Aspiration*

I dedicate this book to my children: *Shauna, Brandon, and Courtney*.

Know that my love goes with you, wherever your aspirations may lead you.

This page is intentionally left blank.



CHAPTER

1

Catalysts for Change

A tourist came in from Orbitville,
parked in the air, and said:
The creatures of this star
are made of metal and glass.
Through the transparent parts
you can see their guts.
Their feet are round and roll
on diagrams of long
measuring tapes, dark
with white lines.
They have four eyes.
The two in back are red.
Sometimes you can see a five-eyed
one, with a red eye turning
on the top of his head.
He must be special—
the others respect him
and go slow
when he passes, winding
among them from behind.
They all hiss as they glide,
like inches, down the marked
tapes. Those soft shapes,
shadowy inside
the hard bodies—are they
their guts or their brains?

—MAY SWENSON, "Southbound on the Freeway"¹

1. Copyright © 1963 by the Literary Estate of May Swenson. Reprinted by permission.

1.1 Introduction

MOST OF US TAKE TECHNOLOGICAL CHANGE FOR GRANTED. In the past two decades alone, we have witnessed the emergence of exciting new technologies, including smartphones, MP3 players, digital photography, email, and the World Wide Web. There is good reason to say we are living in the Information Age. Never before have so many people had such easy access to information. The two principal catalysts for the Information Age have been low-cost computers and high-speed communication networks (Figure 1.1). Even in a society accustomed to change, the rate at which computers and communication networks have transformed our lives is breathtaking.

In 1950 there were no more than a handful of electronic digital computers in the world. Today we are surrounded by devices containing embedded computers. We rely upon microprocessors to control our heating and cooling systems, microwaves, smartphones, elevators, and a multitude of other devices we use every day. Thanks to microprocessors, our automobiles get better gas mileage and produce less pollution. On the other hand, the days of the do-it-yourself tune-up are gone. It takes a mechanic with computerized diagnostic equipment to work on a modern engine.



FIGURE 1.1 Low-cost computers and high-speed communication networks make possible the products of the Information Age, such as the Samsung Galaxy S4. It functions as a phone, email client, Web browser, camera, video recorder, digital compass, and more. (Marian Stanca/Alamy)

In 1990 few people other than college professors used email. Today more than a billion people around the world have email accounts. Email messages are routed instantaneously for very low cost, which can be both a blessing and a curse. Business communications have never been so efficient, but it's not unusual to hear businesspeople complain that they can never get caught up with their email.

The World Wide Web was still being designed in 1990; today it contains more than a trillion pages and makes possible extraordinarily valuable information retrieval systems. Even grade school children are expected to gather information from the Web when writing their reports. However, many parents worry that their Web-surfing children may be exposed to pornographic images or other inappropriate material.

May Swenson has vividly described our ambivalent feelings toward technology. In her poem "Southbound on the Freeway," an alien hovers above an expressway and watches the cars move along [1]. The alien notes "soft shapes" inside the automobiles and wonders, "are they their guts or their brains?" It's fair to ask: Do we drive technology, or does technology drive us?

Our relationship with technology is complicated. We create technology and choose to adopt it. However, once we have adopted a technological device, it can transform us and how we relate to other people and our environment.

Some of the transformations are physical. The neural pathways and synapses in our brains literally change with our experiences. One well-known brain study focused on London taxi drivers. In order to get a license, aspiring London taxi drivers must spend two to four years memorizing the complicated road network of 25,000 streets within 10 kilometers of the Charing Cross train station, as well as the locations of thousands of tourist destinations. The hippocampus is the region of the brain responsible for long-term memory and spatial navigation. Neuroscientists at University College London found that the brains of London taxi drivers have larger-than-average hippocampi and that the hippocampi of aspiring taxi drivers grow as they learn the road network [2].

Stronger longer-term memory and spatial navigation skills are great outcomes of mental exercise, but sometimes the physical effects of our mental exertions are more insidious. For example, studies with macaque monkeys suggest that when we satisfy our hunger for quick access to information through our use of Web browsers, Twitter, and texting, neurons inside our brains release dopamine, producing a desire to seek out additional information, causing further releases of dopamine, and so on, which may explain why we find it difficult to break away from these activities [3, 4].

Adopting a technology can change our perceptions, too. More than 90 percent of cell phone users report that having a cell phone makes them feel safer, but once people get used to carrying a cell phone, losing the phone may make them feel more vulnerable than they ever did before they began carrying one. A Rutgers University professor asked his students to go without their cell phones for 48 hours. Some students couldn't do it. A female student reported to the student newspaper, "I felt like I was going to get raped if I didn't have my cell phone in my hand." Some parents purchase cell phones for their children so that a child may call a family member in an emergency. However, parents who provide a cell phone "lifeline" may be implicitly communicating to their children the idea that people in trouble cannot expect help from strangers [5].



FIGURE 1.2 The Amish carefully evaluate new technologies, choosing those that enhance family and community solidarity. (AP photo/*The Indianapolis Star and News*, Mike Fender)

The Amish understand that the adoption of a new technology can affect the way people relate to each other (Figure 1.2). Amish bishops meet twice a year to discuss matters of importance to the church, including whether any new technologies should be allowed. Their discussion about a new technology is driven by the question, “Does it bring us together, or draw us apart?” You can visit an “Old Order” Amish home and find a gas barbecue on the front porch but no telephone inside, because they believe gas barbecues bring people together while telephones interfere with face-to-face conversations [6].

New technologies are adopted to solve problems, but they often create problems, too. The automobile has given people the ability to travel where they want, when they want. On the other hand, millions of people spend an hour or more each day stuck in traffic commuting between home and work. Refrigerators make it possible for us to keep food fresh for long periods of time. We save time because we don’t have to go grocery shopping every day. Unfortunately, Freon leaking from refrigerators has contributed to the depletion of the ozone layer that protects us from harmful ultraviolet rays. New communication technologies have made it possible for us to get access to news and entertainment from around the world. However, the same technologies have enabled major software companies to move thousands of jobs to India, China, and Vietnam, putting downward pressure on the salaries of computer programmers in the United States [7].

We may not be able to prevent a new technology from being invented, but we do have control over whether to adopt it. Nuclear power is a case in point. Nuclear power plants create electricity without producing carbon dioxide emissions, but they also produce radioactive waste products that must be safely stored for 100,000 years.

Although nuclear power technology is available, no new nuclear power plants were built in the United States for more than 25 years after the accident at Three Mile Island in 1979 [8].

Finally, we *can* influence the rate at which new technologies are developed. Some societies, such as the United States, have a history of nurturing and exploiting new inventions. Congress has passed intellectual property laws that allow people to make money from their creative work, and the federal income tax structure allows individuals to accumulate great wealth.

Most of us appreciate the many beneficial changes that technology has brought into our lives. In health care alone, computed tomography (CT) and magnetic resonance imaging (MRI) scanners have greatly improved our ability to diagnose major illnesses; new vaccines and pharmaceuticals have eradicated some deadly diseases and brought others under control; and pacemakers, hearing aids, and artificial joints have improved the physical well-being of millions.

To sum up, societies develop new technologies to solve problems or make life better, but the use of new technologies changes social conditions and may create new problems. That doesn't mean we should never adopt a new technology, but it does give us a good reason why we should be making informed decisions, weighing the benefits and potential harms associated with the use of new devices. To that end, this book will help you gain a better understanding of contemporary ethical issues related to the use of information technology.

This discussion sets the stage for the remainder of the book. Electronic digital computers and high-performance communication networks are central to contemporary information technology. While the impact of these inventions has been dramatic in the past few decades, their roots go back hundreds of years. Section 1.2 tells the story of the development of computers, showing how they evolved from simple manual calculation aids to complex microprocessors. In Section 1.3 we describe two centuries of progress in networking technology, starting with the semaphore telegraph and culminating in the creation of an email system connecting over a billion users. Section 1.4 shows how information storage and retrieval evolved from the creation of the Greek alphabet to Google. Finally, Section 1.5 discusses some of the moral issues that have arisen from the deployment of information technology.

1.2 Milestones in Computing

Calculating devices have supported the development of commercial enterprises, governments, science, and weapons. As you will see in this section, the introduction of new technologies has often had a social impact.

1.2.1 Aids to Manual Calculating

Adding and subtracting are as old as commerce and taxes. Fingers and toes are handy calculation aids, but to manipulate numbers above 20, people need more than their

own digits. The tablet, the abacus, and mathematical tables are three important aids to manual calculating [9].

Simply having a tablet to write down the numbers being manipulated is a great help. In ancient times, erasable clay and wax tablets served this purpose. By the late Middle Ages, Europeans often used erasable slates. Paper tablets became common in the nineteenth century, and they are still popular today.

An **abacus** is a computing aid in which a person performs arithmetic operations by sliding counters along rods, wires, or lines. The first abacus was probably developed in the Middle East more than two thousand years ago. In a Chinese, Japanese, or Russian abacus, counters move along rods or wires held in a rectangular frame. Beginning in medieval Europe, merchants performed their calculations by sliding wooden or metal counters along lines drawn in a wooden counting board (Figure 1.3). Eventually the word “counter” came to mean not only the disk being manipulated but also the place in a store where transactions take place [9].

Mathematical tables have been another important aid to manual computing for about two thousand years. A great breakthrough occurred in the early seventeenth century, when John Napier and Johannes Kepler published tables of logarithms. These tables were tremendous time savers to anyone doing complicated math because they allowed them to multiply two numbers by simply adding their logarithms. Many other useful tables were created as well. For example, businesspeople consulted tables to compute interest and convert between currencies. Today, people who compute their income taxes “by hand” make use of tax tables to determine how much they owe.

Even with tablets, abacuses, and mathematical tables, manual calculating is slow, tedious, and error-prone. To make matters worse, mathematical tables prepared centuries ago usually contained errors. That’s because somebody had to compute each table entry and somebody had to typeset each entry, and errors could occur in either of these steps. Advances in science, engineering, and business in the post-Renaissance period motivated European inventors to create new devices to make calculations faster and more reliable and to automate the printing of mathematical tables.

1.2.2 Mechanical Calculators

Blaise Pascal had a weak physique but a powerful mind. When he got tired of summing by hand long columns of numbers given him by his father, a French tax collector, he constructed a mechanical calculator to speed the chore. Pascal’s calculator, built in 1640, was capable of adding whole numbers containing up to six digits. Inspired by Pascal’s invention, the German Gottfried Leibniz constructed a more sophisticated calculator that could add, subtract, multiply, and divide whole numbers. The hand-cranked machine, which he called the Step Reckoner, performed multiplications and divisions through repeated additions and subtractions, respectively. The calculators of Pascal and Leibniz were not reliable, however, and did not enjoy commercial success.

In the nineteenth century, advances in machine tools and mass-production methods, combined with larger markets, made possible the creation of practical calculating machines. Frenchman Charles Thomas de Colmar utilized the stepped drum gear mech-



FIGURE 1.3 This illustration from Gregor Reisch's *Margarita Philosophica*, published in 1503, shows two aids to manual calculating. The person on the left is using a tablet; the person on the right is adding numbers using a counting board, a type of abacus. (Heritage Images/Corbis)

anism invented by Leibniz to create the Arithmometer, the first commercially successful calculator. Many insurance companies purchased Arithmometers to help their actuaries compute rate tables more rapidly [9].

Swedish publisher Georg Scheutz was intimately familiar with printing errors associated with the production of mathematical tables. He resolved to build a machine capable of automatically calculating and typesetting table values. Scheutz knew about the earlier work of English mathematician Charles Babbage, who had demonstrated how a machine could compute the values of polynomial functions through the method of

differences. Despite promising early results, Babbage's efforts to construct a full-scale Difference Engine had been unsuccessful. In contrast, Georg Scheutz and his son Edward, who developed their own designs, completed the world's first printing calculator: a machine capable of calculating mathematical tables and typesetting the values onto molds. The Dudley Observatory in Albany, New York, purchased the Scheutz difference engine in 1856. With support from the US Nautical Almanac Office, astronomers used the machine to help them compute the motion of Mars and the refraction of starlight. Difference engines were never widely used; the technology was eclipsed by the emergence of simpler and less expensive calculating machines [9].

America in the late 1800s was fertile ground for the development of new calculating technologies. This period of American history, commonly known as the Gilded Age, was characterized by rapid industrialization, economic expansion, and a concentration of corporate power. Corporations merged to increase efficiency and profits, but the new, larger corporate organizations had multiple layers of management and multiple locations. In order for middle- and upper-level managers to monitor and improve performance, they needed access to up-to-date, comprehensive, reliable, and affordable information. All these requirements could not be met by bookkeepers and accountants using pen and paper to sum long columns of transactions by hand [10].

To meet this demand, many entrepreneurs began producing adding and calculating machines. One of these inventors was William Burroughs, a former bank clerk who had spent long days adding columns of figures. Burroughs devised a practical adding machine and offered it for sale. He found himself in a cutthroat market; companies competed fiercely to reduce the size of their machines and make them faster and easier to use. Burroughs distinguished himself from his competitors by putting together first-class manufacturing and marketing organizations, and by the 1890s the Burroughs Adding Machine Company led the industry. Calculating machines were entrenched in the offices of large American corporations by the turn of the century [10].

The adoption of mechanical calculators led to the "de-skilling" and "feminization" of bookkeeping (Figure 1.4). Before the introduction of calculating machines, offices were a male bastion, and men who could rapidly compute sums by hand were at a premium. Calculators leveled the playing field, making people of average ability quite productive. In fact, a 1909 Burroughs study concluded that a clerk using a calculator was six times faster than a clerk adding the same column of figures by hand [11]. As managers introduced mechanical calculators into offices, they replaced male bookkeepers with female bookkeepers and lowered wages. In 1880 only 5.7 percent of bookkeepers, cashiers, and accountants were women, but by 1910 the number of women in these jobs had risen to 38.5 percent [12].

1.2.3 Cash Register

Store owners in the late 1800s faced challenges related to accounting and embezzlement. Keeping accurate sales records was becoming more difficult as smaller stores evolved into "department stores" with several departments and many clerks. Preventing embezzlement was tricky when clerks could steal cash simply by not creating receipts for some sales.



FIGURE 1.4 Mechanical calculators led to the “de-skilling” and “feminization” of book-keeping. (Underwood Archives/Getty Images)

While on a European holiday in 1878, Ohio restaurateur James Ritty saw a mechanical counter connected to the propeller shaft of his ship. A year later he and his brother John used that concept to construct the first cash register, essentially an adding machine capable of expressing values in dollars and cents. Enhancements followed rapidly, and by the early 1900s the cash register had become an important information-processing device (Figure 1.5). Cash registers created printed, itemized receipts for customers, maintained printed logs of transactions, and performed other accounting functions that provided store owners with the detailed sales records they needed.

Cash registers also made embezzlement by clerks more difficult. The bell made it impossible for clerks to sneak money from the cash drawer and helped ensure that every sale was “rung up.” Printed logs made it easy for department store owners to compare cash on hand against sales receipts [10].

1.2.4 Punched-Card Tabulation

As corporations and governmental organizations grew larger in the late 1800s, they needed to handle greater volumes of information. One of these agencies was the US Bureau of the Census, which collected and analyzed information on tens of millions of residents every decade. Aware of the tedium and errors associated with clerks manually copying and tallying figures, several Census Bureau employees developed mechanical



FIGURE 1.5 An NCR cash register in Miller's Shoe Shine Parlor, Dayton, Ohio (1904). (The NCR Archive at Dayton History)

tabulating machines. Herman Hollerith created the most successful device. Unlike a predecessor, who chose to record information on rolls of paper, Hollerith decided to record information on punched cards. The use of punched cards to store data was a much better approach because cards could be sorted into groups, allowing the computation of subtotals by categories. Hollerith's equipment proved to be a great success when used in the 1890 census. In contrast to the 1880 census, which had required eight years to complete, the 1890 census was finished in only two years. Automating the census saved the Census Bureau five million dollars, about one-third of its annual budget [13].

Other data-intensive organizations found applications for punched cards. Railroads used them to improve their accounting operations and send bills out more frequently. Retail organizations, such as Marshall Field's, used punched cards to perform more sophisticated analyses of information generated by the cash registers at its many stores. The Pennsylvania Steel Company and other heavy industries began to use punched-card technology to do cost accounting on manufacturing processes.

The invention of sorters, tabulators, and other devices to manipulate the data on punched cards created a positive feedback loop. As organizations began using tabulating machines, they thought up new uses of information-processing equipment, stimulating further technological innovations.

International Business Machines (IBM) is the corporate descendant of Hollerith's company. Over a period of several decades, IBM and its principal competitor, Remington Rand, developed sophisticated machines based around punched cards: card

punches, card verifiers, card tabulators, card sorters, and alphabetizers. Users used these devices to create **data-processing systems** that received input data, performed one or more calculations, and produced output data. Within these systems, punched cards stored input data, intermediate results, and output data. In the most complicated systems, punched cards also stored the program—the steps of the computational process to be followed. Early systems relied on human operators to carry cards from one machine to the next. Later systems had electrical connections that allowed the output of one machine to be transmitted to the next machine without the use of punched cards or human intervention.

Organizations with large data-processing needs found punched-card tabulators and calculators to be valuable devices, and they continually clamored for new features that would improve the computational capabilities and speed of their systems [10]. These organizations would become a natural market for commercial electronic digital computers.

IBM machines played an infamous role in the Holocaust. After Adolf Hitler came to power in Germany in 1933, IBM chief executive Thomas J. Watson overlooked well-publicized accounts of anti-Semitic violence and the opening of concentration camps, focusing instead on a golden business opportunity. The firm expanded the operations of its German subsidiary, Dehomag, built a new factory in Germany, and actively sought business from the German government. Tabulating, sorting, collating, and alphabetizing machines and support services provided by Dehomag allowed the Nazi government to rapidly conduct censuses, identify acknowledged Jews and those with Jewish ancestors, and generate the alphabetical lists of names needed to efficiently seize their assets, confine them to ghettos, and deport them to death camps [13].

1.2.5 Precursors of Commercial Computers

Several computing devices developed during and immediately after World War II paved the way for the commercialization of electronic digital computers.

Between 1939 and 1941, Iowa State College professor John Atanasoff and his graduate student Clifford Berry constructed an electronic device for solving systems of linear equations. The Atanasoff-Berry Computer was the first computing device built with vacuum tubes, but it was not programmable.

Dr. John W. Mauchly, a physics professor at the University of Pennsylvania, visited Iowa State College in 1941 to learn more about the Atanasoff-Berry Computer. After he returned to Penn, Mauchly worked with J. Presper Eckert to create a design for an electronic computer to speed the computation of artillery tables for the US Army. They led a team that completed work on the ENIAC (electronic numerical integrator and computer) in 1946. As it turns out, the war ended before the ENIAC could provide the Army with any ballistics tables, but its speed was truly impressive. A person with a desk calculator could compute a 60-second trajectory in 20 hours. The ENIAC performed the computation in 30 seconds. In other words, the ENIAC was 2,400 times faster than a person with a desk calculator.

The ENIAC had many features of a modern computer. All of its internal components were electronic, and it could be programmed to perform a variety of computations. However, its program was not stored inside memory. Instead, it was “wired in”



FIGURE 1.6 The ENIAC's first six programmers were women. Every instruction was programmed by connecting several wires into plugboards. (© Corbis)

from the outside. Reprogramming the computer meant removing and reattaching many wires. This process could take many days (Figure 1.6).

Even before the ENIAC was completed, work began on a follow-on system called the EDVAC (electronic discrete variable automatic computer). The design of the EDVAC incorporated many improvements over the ENIAC. The most important improvement was that the EDVAC stored the program in primary memory, along with the data manipulated by the program. In 1946 Eckert, Mauchly, and several other computer pioneers gave a series of 48 lectures at the Moore School. While some of the lectures discussed lessons learned from the ENIAC, others focused on the design of its successor, the EDVAC. These lectures influenced the design of future machines built in the United States and the United Kingdom.

During World War II, British engineer F. C. Williams was actively involved in the development of cathode ray tubes (CRTs) used in radar systems. After the war, he decided to put his knowledge to use by figuring out how to use a CRT as a storage device for digital information. In early 1948 a team at the University of Manchester set out to build a small computer that would use a CRT storage device, now called the Williams Tube, to store the program and its data. They called their system the Small-Scale Experimental Machine. The computer successfully executed its first program in 1948. The Small-Scale Experimental Machine was the first operational, fully electronic computer system that had both program and data stored in its memory.



FIGURE 1.7 CBS news coverage of the 1952 presidential election included predictions made by a UNIVAC computer. When the computer predicted Eisenhower would win in a landslide, consternation followed. (Hagley Museum and Library, accession number 1984.240)

1.2.6 First Commercial Computers

In 1951 the British corporation Ferranti Ltd. introduced the Ferranti Mark 1, the world's first commercial computer. The computer was the direct descendant of research computers constructed at the University of Manchester. Ferranti delivered nine computers between 1951 and 1957, and later Ferranti models boasted a variety of technological breakthroughs, thanks to the company's close association with research undertaken at the University of Manchester and Cambridge University.

After completing work on the ENIAC, Eckert and Mauchly formed their own company to produce a commercial digital computer. The Eckert-Mauchly Computer Corporation signed a preliminary agreement with the National Bureau of Standards (representing the Census Bureau) in 1946 to develop a commercial computer, which they called the UNIVAC, for universal automatic computer. The project experienced huge cost overruns, and by 1950 the Eckert-Mauchly Computer Corporation was on the brink of bankruptcy. Remington Rand bought them out and delivered the UNIVAC I to the US Bureau of the Census in 1951 [14].

In a public relations coup, Remington Rand cooperated with CBS to use a UNIVAC computer to predict the outcome of the 1952 presidential election (Figure 1.7). The events of election night illustrate the tough decisions people can face when computers produce unexpected results.

Adlai Stevenson had led Dwight Eisenhower in polls taken before the election, but less than an hour after the polls closed, with just 7 percent of the votes tabulated, the UNIVAC was predicting Dwight Eisenhower would win the election in a landslide. When CBS correspondent Charles Collingwood asked Remington Rand for the computer's prediction, however, he was given the run-around. The computer's engineers were convinced there was a programming error. For one thing, UNIVAC was predicting that Eisenhower would carry several Southern states, and everybody "knew" that Republican presidential candidates never won in the South. Remington Rand's director of advanced research ordered the engineers to change the programming so the outcome would be closer to what the political pundits expected. An hour later, the reprogrammed computer predicted that Eisenhower would win by only nine electoral votes, and that's what CBS announced. As it turns out, the computer was right and the human "experts" were wrong. Before being reprogrammed, UNIVAC had predicted Eisenhower would win 438 electoral votes to 93 for Stevenson. The official result was a 442–89 victory for Eisenhower [14].

In America in the early 1950s, the word "UNIVAC" was synonymous with "computer." Remington Rand sold a total of 46 UNIVACs to government agencies, such as the US Air Force, the US Army Map Service, the Atomic Energy Commission, and the US Navy, as well as to large corporations and public utilities, such as General Electric, Metropolitan Life, US Steel, Du Pont, Franklin Life Insurance, Westinghouse, Pacific Mutual Life Insurance, Sylvania Electric, and Consolidated Edison.

Office automation leader IBM did not enter the commercial computer market until 1953, and its initial products were inferior to the UNIVAC. However, IBM quickly turned the tables on Remington Rand, thanks to a larger base of existing customers, a far superior sales and marketing organization, and a much greater investment in research and development. In 1955 IBM held more than half the market, and by the mid-1960s IBM dominated the computer industry with 65 percent of total sales, compared to 12 percent for number two computer maker Sperry Rand (the successor to Remington Rand) [14].

1.2.7 Programming Languages and Time-Sharing

In the earliest digital computers, every instruction was coded as a long string of 0s and 1s. People immediately began looking for ways to make coding faster and less error-prone. One early improvement was the creation of assembly language, which allowed programmers to work with symbolic representations of the instruction codes. Still, one assembly language instruction was required for every machine instruction. Programmers wanted fewer, higher-level instructions to generate more machine instructions. In 1951 Frances Holberton, one of the six original ENIAC programmers, created a sort-merge generator for the UNIVAC that took a specification of files to be manipulated and automatically produced the machine program to do the sorting and merging. Building on this work, Grace Murray Hopper, also at Remington Rand, developed the A-0 system that automated the process of linking together subroutines to form the complete machine code [15].

Over at IBM, John Backus convinced his superiors of the need for a higher-level programming language for IBM computers. He led the effort to develop the IBM Mathematical Formula Translating System, or FORTRAN. Designed for scientific applications, the first system was completed in 1957. Many skeptics believed that any “automatic programming” system would generate inefficient machine code compared to hand-coded assembly language, but they were proven wrong: the FORTRAN compiler generated high-quality code. What’s more, programmers could write FORTRAN programs 5 to 20 times faster than the equivalent assembly language programs. Most programmers quickly shifted allegiance from assembly language to FORTRAN. Eventually other computer manufacturers developed their own FORTRAN compilers, and FORTRAN became an international standard [16].

Meanwhile, business-oriented programming languages were also being developed by several computer manufacturers. Grace Murray Hopper specified FLOW-MATIC, an English-like programming language for the UNIVAC. Other manufacturers began to develop their own languages. Customers didn’t like incompatible languages, because it meant programs written for one brand of computer had to be rewritten before they could be run on another brand of computer. In 1959 an extremely important customer, the US Department of Defense, brought together a committee to develop a common business-oriented programming language that all manufacturers would support. The committee wrote the specification for COBOL. By requiring manufacturers to support COBOL in order to get defense contracts, the US Department of Defense helped ensure its widespread adoption [17].

In the early 1960s, John Kemeny and Thomas Kurtz at Dartmouth College directed teams of undergraduate students who developed a time-sharing system and an easy-to-learn programming language. The Dartmouth Time-Sharing System (DTSS) gave multiple people the ability to edit and run their programs simultaneously, by dividing the computer’s time among all the users. Time-sharing made computers accessible to more people because it allowed the cost of owning and operating a computer system to be divided among a large pool of users who purchased the right to connect to the system [18].

The development of BASIC, a simple, easy-to-learn programming language, was another important step toward making computers accessible to a wider audience. Kemeny and Kurtz saw BASIC as a way to teach programming, and soon many other educational institutions began teaching students how to program using Dartmouth BASIC. The language’s popularity led computer manufacturers to develop their own versions of BASIC [18].

1.2.8 Transistor and Integrated Circuit

Although the British had radar installations at the beginning of World War II, it became clear during the Battle of Britain that their systems were inadequate. The British and Americans worked together to develop microwave radar systems capable of locating enemy planes more precisely. Microwave radar required higher-frequency receivers utilizing semiconductors, and in the process of manufacturing microwave radar systems



FIGURE 1.8 The eight founders of Fairchild Semiconductor on the factory floor. Gordon Moore is second from the left and Robert Noyce is on the right. (Wayne Miller/Magnum Photos, Inc.)

for the war effort, several American companies, including AT&T, greatly improved their ability to create semiconductors [19].

AT&T was on the lookout for a new technology to replace the vacuum tube. Its long-distance network relied on vacuum tubes to amplify signals, but the tubes required a lot of power, generated a lot of heat, and burned out like lightbulbs. After the war, AT&T put together a team of Bell Labs scientists, led by Bill Shockley, to develop a semiconductor substitute for the vacuum tube. In 1948 Bell Labs announced the invention of such a device, which they called the **transistor** [20].

While most electronics companies ignored the invention of the transistor, Bill Shockley understood its potential. He left Bell Labs and moved to Palo Alto, California, where he founded Shockley Semiconductor in 1956. He hired an exceptional team of engineers and physicists, but many disliked his heavy-handed management style [20]. In September 1957, eight of Shockley's most talented employees, including Gordon Moore and Robert Noyce, walked out. The group, soon to be known as the "traitorous eight," founded Fairchild Semiconductor (Figure 1.8). By this time transistors were being used in a wide variety of devices, from transistor radios to computers. While transistors were far superior to vacuum tubes, they were still too big for some applications. Fairchild Semiconductor set out to produce a single semiconductor device containing transistors, capacitors, and resistors; in other words, an **integrated circuit**. Another firm, Texas Instruments, was on the same mission. Today, Robert Noyce of Fairchild Semiconduc-

tor and Jack Kilby of Texas Instruments are credited for independently inventing the integrated circuit [21].

The Cold War between the United States and the Soviet Union played an important role in advancing integrated circuit technology. American engineers developing the Minuteman II ballistic missile in the early 1960s decided to use integrated circuits to improve the processing speed of its guidance computer. The Minuteman II program was the single largest consumer of integrated circuits in the United States between 1962 and 1965, representing about 20 percent of total sales. During these years companies learned how to make rugged, reliable integrated circuits [9]. They also continued to shrink the components within the integrated circuits, leading to an exponential increase in their power. Gordon Moore noted this trend in a 1965 paper and predicted it would continue. Today **Moore's law** refers to the phenomenon of integrated circuits becoming twice as powerful roughly every two years.

1.2.9 IBM System/360

The integrated circuit made possible the construction of much more powerful and reliable computers. The 1960s was the era of mainframe computers—large computers designed to serve the data-processing needs of large businesses. Mainframe computers enabled enterprises to centralize all of their data-processing applications in a single system. As we have seen, by this time IBM dominated the mainframe market in the United States.

In 1964 IBM announced the System/360, a series of 19 compatible computers with varying levels of computing speed and memory capacity (Figure 1.9). Because the systems were software compatible, a business could upgrade its computer without having to rewrite its application programs. This feature was important, because by the 1960s companies were making much larger investments in software.

1.2.10 Microprocessor

In 1968 Robert Noyce and Gordon Moore left Fairchild Semiconductor to found another semiconductor manufacturing company, which they named Intel. A year later Japanese calculator manufacturer Busicom approached Intel about designing 12 custom chips for use in a new scientific calculator. Intel agreed to provide the chips and assigned responsibility for the project to Marcian “Ted” Hoff. After reviewing the project, Hoff suggested that it was not in Intel's best interest to manufacture a custom chip for every customer. As an alternative, he suggested that Intel create a general-purpose chip that could be programmed to perform a wide variety of tasks. Each customer could program the chip to meet its particular needs. Intel and Busicom agreed to the plan, which reduced the required number of chips for Busicom's calculator from 12 to 4. A year of development by Ted Hoff, Stanley Mazor, and Federico Faggin led to the release of the Intel 4004, the world's first **microprocessor**. Inside the 1/8-inch $\times$ 1/6-inch chip were 2,300 transistors, giving the Intel 4004 the same computing power as the ENIAC, which had occupied 3,000 cubic feet.

Microprocessors made it possible to integrate computers into everyday devices. Today we're surrounded by devices containing microprocessors: cell phones, MP3 players,



FIGURE 1.9 In the 1960s, IBM dominated the mainframe computer market in the United States. (IBM/AP Photo)

digital cameras, wristwatches, ATM machines, automobiles, microwave ovens, thermostats, traffic lights, and much more. The highest-profile use of microprocessors, however, is in personal computers.

1.2.11 Personal Computer

During the Vietnam conflict in the late 1960s and early 1970s, the area around San Francisco was home to a significant counterculture, including a large number of antiwar and antiestablishment activists. The do-it-yourself idealism of the power-to-the-people movement intersected with advances in computer technology in a variety of ways, including the *Whole Earth Catalog*, the People's Computer Company, and the Homebrew Computer Club [22].

The *Whole Earth Catalog*, first published in 1968, was, in the words of Steve Jobs, “sort of like Google in paperback form” [23]—an effort to pull together in a single

large volume lists of helpful tools, in this case for the creation of a more just and environmentally sensitive society. The definition of “tools” was broad; the catalog’s lists included books, classes, garden tools, camping equipment, and (in later issues) early personal computers. “With the *Whole Earth Catalog*, Stewart Brand offered a generation of computer engineers and programmers an alternative vision of technology as a tool for individual and collective transformation” [24, p. 104].

The People’s Computer Company was a not-for-profit corporation dedicated to educating people on how to use computers. One of its activities was publishing a newspaper. The cover of the first issue read: “Computers are mostly used against people instead of for people, used to control people instead of to free them, time to change all that—we need a PEOPLE’S COMPUTER COMPANY” [25]. Typical issues contained programming tips and the source code to programs, particularly educational games written in BASIC. The newspaper’s publisher, Bob Albrecht, said, “I was heavily influenced by the *Whole Earth Catalog*. I wanted to give away ideas” [24, p. 114]. The People’s Computer Company also set up the People’s Computer Center in a strip mall in Menlo Park. The center allowed people to rent teletype terminals connected to a time-shared computer. A large number of teenagers were drawn to computing through Friday evening game-playing sessions. Many users wrote their own programs, and the center promoted a culture in which computer enthusiasts freely shared software with each other.

In 1975 the Homebrew Computer Club, an outgrowth of the People’s Computer Company, became a meeting place for hobbyists interested in building personal computers out of microprocessors. A company in Albuquerque, New Mexico, called MITS, had recently begun shipping the Altair 8800 personal computer, and during the first few Homebrew Computer Club meetings, members showed off various enhancements to the Altair 8800. Progress was frustratingly slow, however, due to the lack of a higher-level programming language.

Three months after the establishment of the Homebrew Computer Club, MITS representatives visited Palo Alto to demonstrate the Altair 8800 and the BASIC interpreter created by Paul Allen and Bill Gates, who had a tiny company called Micro-Soft. The audience in the hotel conference room was far larger than expected, and during the overcrowded and chaotic meeting somebody acquired a paper tape containing the source code to Altair BASIC. More than 70 copies of the tape were handed out at the next meeting of the Homebrew Computer Club. After that, free copies of the interpreter proliferated. Some hobbyists felt that the asking price of \$500 for the BASIC interpreter was too high, considering that the Altair computer itself cost only \$395 as a kit or \$495 preassembled [22].

Bill Gates responded by writing “An Open Letter to Hobbyists” that was reprinted in a variety of publications. In the letter he asserted that less than 10 percent of all Altair owners had purchased BASIC, even though far more people than that were using it. According to Gates, the royalties Micro-Soft had received from Altair BASIC made the time spent on the software worth less than \$2 an hour. He wrote, “Nothing would please me more than being able to hire ten programmers and deluge the hobby market with good software,” but the theft of software created “very little incentive” for his company to release new products [22].



FIGURE 1.10 Steve Jobs (right) convinced Steve Wozniak they should go into business selling the personal computer Wozniak designed. They named their company Apple Computer.
(© Kimberly White/Corbis)

The controversy over Altair BASIC did not slow the pace of innovations. Hobbyists wanted to do more than flip the toggle switches and watch the lights blink on the Altair 8800. Steve Wozniak, a computer engineer at Hewlett-Packard, created a more powerful personal computer that supported keyboard input and television monitor output. Wozniak's goal was to make a machine for himself and to impress other members of the Homebrew Computer Club, but his friend Steve Jobs thought of a few improvements and convinced Wozniak they should go into business (Figure 1.10). They raised \$1,300 by selling Jobs's Volkswagen van and Wozniak's Hewlett-Packard scientific calculator, launching Apple Computer. Although the company sold only 200 Apple I computers, its next product, the Apple II, became one of the most popular personal computers of all time.

By the end of the 1970s, many companies, including Apple Computer and Tandy, were producing personal computers. While hundreds of thousands of people bought personal computers for home use, businesses were reluctant to move to the new computer platform. However, two significant developments made personal computers more attractive to businesses.

The first development was the computer spreadsheet program. For decades firms had used spreadsheets to perform financial predictions. Manually computing spreadsheets was monotonous and error-prone, since changing a value in a single cell could require updating many other cells. In the fall of 1979, Bob Frankston and Harvard MBA

student Dan Bricklin released their program, called VisiCalc, for the Apple II. VisiCalc's labor-saving potential was obvious to businesses. After a slow start, it quickly became one of the most popular application programs for personal computers. Sales of the Apple II computer increased significantly after the introduction of VisiCalc.

The second development was the release of the IBM PC in 1981. The IBM name exuded reliability and respectability, making it easier for companies to make the move to desktop systems for their employees. As the saying went, "Nobody ever got fired for buying from IBM." In contrast to the approach taken by Apple Computer, IBM decided to make its PC an open architecture, meaning the system was built from off-the-shelf parts and other companies could manufacture "clones" with the same functionality. This decision helped to make the IBM PC the dominant personal computer architecture.

The success of IBM-compatible PCs fueled the growth of Microsoft. In 1980 IBM contracted with Microsoft to provide the DOS operating system for the IBM PC. Microsoft let IBM have DOS for practically nothing, but in return IBM gave Microsoft the right to collect royalties from other companies manufacturing PC-compatible computers. Microsoft profited handsomely from this arrangement when PC-compatibles manufactured by other companies gained more than 80 percent of the PC market [26].

1.3 Milestones in Networking

In the early nineteenth century, the United States fell far behind Europe in networking technology. The French had begun constructing a network of telegraph towers in the 1790s, and forty years later there were towers all over the European continent (Figure 1.11). At the top of each tower was a pair of semaphores. Operators raised and lowered the semaphores; each pattern corresponded to a letter or symbol. A message initiated at one tower would be seen by another tower within viewing distance. The receiving tower would then repeat the message for the next tower in the network, and so on. This optical telegraph system could transmit messages at the impressive rate of about 350 miles per hour when skies were clear.

In 1837 Congress asked for proposals to create a telegraph system between New York and New Orleans. It received one proposal based on proven European technology. Samuel Morse submitted a radically different proposal. He suggested constructing a telegraph system that used electricity to communicate the signals. Let's step back and review some of the key discoveries and inventions that enabled Morse to make his dramatic proposal.

1.3.1 Electricity and Electromagnetism

Amber is a hard, translucent, yellowish-brown fossil resin often used to make beads and other ornamental items. About 2,600 years ago the Greeks discovered that if you rub amber, it becomes charged with a force enabling it to attract light objects such as feathers and dried leaves. The Greek word for amber is *ηλεκτρον* (electron). Our word "electric" literally means "to be like amber."



FIGURE 1.11 A semaphore telegraph tower on the first line from Paris to Lille (1794). (Leemage/Getty Image)

For more than two thousand years amber's ability to attract other materials was seen as a curiosity with no practical value, but in the seventeenth and eighteenth centuries scientists began to study electricity in earnest. Alessandro Volta, a professor of physics at the University of Pavia, made a key breakthrough when he discovered that electricity could be generated chemically. He produced an electric current by submerging two different metals close to each other in an acid. In 1799 Volta used this principle to create the world's first battery. Volta's battery produced an electric charge more than 1,000 times as powerful as that produced by rubbing amber. Scientists soon put this power to practical use.

In 1820 Danish physicist Christian Oersted discovered that an electric current creates a magnetic field. Five years later British electrician William Sturgeon constructed an electromagnet by coiling wire around a horseshoe-shaped piece of iron. When he ran an electric current through the coil, the iron became magnetized. Sturgeon showed how a single battery was capable of producing a charge strong enough to pick up a nine-pound metal object.

In 1830 American professor Joseph Henry rigged up an experiment that showed how a telegraph machine could work. He strung a mile of wire around the walls of his classroom at the Albany Academy. At one end he placed a battery; at the other end he connected an electromagnet, a pivoting metal bar, and a bell. When Henry connected the battery, the electromagnet attracted the metal bar, causing it to ring the bell. Disconnecting the battery allowed the bar to return to its original position. In this way he could produce a series of rings.



FIGURE 1.12 Pony Express riders lost their jobs when the US transcontinental telegraph line was completed in 1861. (© North Wind Picture Archives/Alamy)

1.3.2 Telegraph

Samuel Morse, a professor of arts and design at New York University, worked on the idea of a telegraph during most of the 1830s, and in 1838 he patented his design of a telegraph machine. The US Congress did not approve Morse's proposal in 1837 to construct a New York-to-New Orleans telegraph system, but it did not fund any of the other proposals either. Morse persisted with his lobbying, and in 1843 Congress appropriated \$30,000 to Morse for the construction of a 40-mile telegraph line between Washington, DC, and Baltimore, Maryland.

On May 1, 1844, the Whig party convention in Baltimore nominated Henry Clay for president. The telegraph line had been completed to Annapolis Junction at that time. A courier hand-carried a message about Clay's nomination from Baltimore to Annapolis Junction, where it was telegraphed to Washington. This was the first news reported via telegraph. The line officially opened on May 24. Morse, seated in the old Supreme Court chamber inside the US Capitol, sent his partner in Baltimore a verse from the Bible: "What hath God wrought?"

The value of the telegraph was immediately apparent, and the number of telegraph lines quickly increased. By 1846 telegraph lines connected Washington, Baltimore, Philadelphia, New York, Buffalo, and Boston. In 1850 twenty different companies operated 12,000 miles of telegraph lines. The first transcontinental telegraph line was completed in 1861, putting the Pony Express out of business (Figure 1.12). The telegraph was the sole method of rapid long-distance communication until 1877. By this time the United States was networked by more than 200,000 miles of telegraph wire [27].

The telegraph was a versatile tool, and people kept finding new applications for it. For example, by 1870 fire alarm telegraphs were in use in 75 major cities in the United

States. New York City alone had 600 fire alarm telegraphs. When a person pulled the lever of the alarm box, it automatically transmitted a message identifying its location to a fire station. These devices greatly improved the ability of fire departments to dispatch equipment quickly to the correct location [27].

1.3.3 Telephone

Alexander Graham Bell was born in Edinburgh, Scotland, into a family focused on impairments of speech and hearing. His father and grandfather were experts in elocution and the correction of speech. His mother was almost completely deaf. Bell was educated to follow in the same career path as his father and grandfather, and he became a teacher of deaf students. Later, he married a deaf woman.

Bell pursued inventing as a means of achieving financial independence. At first he focused on making improvements to the telegraph. A significant problem with early telegraph systems was that a single wire could transmit only one message at a time. If multiple messages could be sent simultaneously along the same wire, communication delays would be reduced, and the value of the entire system would increase.

Bell's solution to this problem was called a harmonic or musical telegraph. If you imagine hearing Morse code, it's obvious that all of the dots and dashes are the same note played for a shorter or longer period of time. The harmonic telegraph assigned a different note (different sound frequency) to each message. At the receiving end, different receivers could be tuned to respond to different notes, as you can tune your radio to hear only what is broadcast by a particular station.

Bell knew that the human voice is made up of sounds at many different frequencies. From his work on the harmonic telegraph, he speculated that it should be possible to capture and transmit human voice over a wire. He and Thomas A. Watson succeeded in transmitting speech electronically in 1876. Soon after, they commercialized their invention.

Nearly all early telephones were installed in businesses. Leasing a telephone was expensive, and most people focused on its commercial value rather than its social value. However, the number of phones placed in homes increased rapidly in the 1890s, after Bell's first patent expired.

Once telephones were placed in the home, the traditional boundaries between private family life and public business life became blurred. People enjoyed being able to conduct business transactions from the privacy of their home, but they also found that a ringing telephone could be an unwelcome interruption [28].

Another consequence of the telephone was that it eroded traditional social hierarchies. An 1897 issue of *Western Electrician* reports that Governor Chauncey Depew of New York was receiving unwanted phone calls from ordinary citizens: "Every time they see anything about him in the newspapers, they call and tell him what a 'fine letter he wrote' or 'what a lovely speech he made,' or ask if this or that report is true; and all this from people who, if they came to his office, would probably never say more than 'Good morning'" [29].

People also worried about the loss of privacy brought about by the telephone. In 1877 the *New York Times* reported that telephone men responsible for operating an early system in Providence, Rhode Island, overheard many confidential conversations. The writer fretted that telephone eavesdropping would make it dangerous for anyone in Providence to accept a nomination for public office [28].

The telephone enabled the creation of the first “online” communities. In rural areas the most common form of phone service was the party line: a single circuit connecting multiple phones to the telephone exchange. Party lines enabled farmers to gather by their phones every evening to talk about the weather and exchange gossip [30].

The power of this new medium was demonstrated in the Bryan/McKinley presidential election of 1896. For the first time, presidential election returns were transmitted directly into people’s homes. “Thousands sat with their ear glued to the receiver the whole night long, hypnotized by the possibilities unfolding to them for the first time” [31].

1.3.4 Typewriter and Teletype

For hundreds of years people dreamed of a device that would allow an individual to produce a document that looked as if it had been typeset, but the dream was not realized until 1867, when Americans Christopher Sholes, Carlos Glidden, and Samuel Soule patented the first typewriter. In late 1873 Remington & Sons Company, famous for guns and sewing machines, produced the first commercial typewriter. It was difficult to use and was not well received; Remington & Co. sold only 5,000 machines in the first five years. However, the typewriter did get the attention of Mark Twain, who used it to produce *Tom Sawyer*, which may have been the world’s first typewritten manuscript. By 1890 more reliable typewriters were being produced, and the typewriter became a common piece of office equipment [32].

In 1908 Charles and Howard Krum succeeded in testing an experimental machine that allowed a modified Oliver typewriter to print a message transmitted over a telegraph line. They called their invention the teletype. During the 1920s, news organizations began using teletype machines to transmit stories between distant offices, and Wall Street firms began sending records of stock transactions over teletypes.

1.3.5 Radio

Earlier we described how the experiments of Oersted, Sturgeon, and Henry led to the development of the electromagnet and the telegraph. The connection between electricity and magnetism remained mysterious, however, until Scottish physicist James Clerk Maxwell published a mathematical theory demonstrating their relationship. This theory predicted the existence of an electromagnetic wave spreading with the velocity of light. It also predicted that light itself was an electromagnetic phenomenon. In 1885 Heinrich Hertz successfully generated electromagnetic waves, proving the correctness of Maxwell’s theory.

Guglielmo Marconi put Hertz's discovery to practical use by successfully transmitting radio signals in the hills outside Bologna, Italy, in 1895. Unable to attract the attention of the Italian government, he took his invention to England, where he founded the Marconi Wireless Telegraph Company. The name of the company reflects Marconi's concept of how his invention would be used. To Marconi, radio, or "wireless," was a superior way to transmit telegraph messages.

David Sarnoff emigrated from Russia to the United States with his family when he was nine. When he had completed school, he landed a position with the Marconi Wireless Telegraph Company. In 1912 Sarnoff made a name for himself when his office relayed news about the sinking of the *Titanic*. Four years later, Sarnoff suggested the use of radio as an entertainment device, writing: "I have in mind a plan of development which would make radio a household utility in the same sense as the piano or phonograph. . . . The receiver can be designed in the form of a simple music box . . . [which] can be placed in the parlor or living room" [33]. In two decades, Sarnoff's vision had become a reality.

The power of radio as a medium of mass communication was demonstrated on the evening of October 30, 1938 (the night before Halloween). From CBS Radio Studio One in New York, Orson Welles and the Mercury Theater put on a one-hour dramatization of H. G. Wells's *War of the Worlds* (Figure 1.13). To increase suspense, the play was performed as a series of news bulletins interrupting a concert of dance music. These bulletins described events occurring on a farm near Grovers Mill, New Jersey. Many listeners panicked. "People packed the roads, hid in cellars, loaded guns, even wrapped their heads in wet towels as protection from Martian poison gas, oblivious to the fact that they were acting out the role of the panic-stricken public that actually belonged in a radio play" [34].

1.3.6 Television

Broadcasting video over a wire began in 1884 with the invention of an electro-mechanical television by Paul Nipkow, but the first completely electronic television transmission was made in 1927 by Philo Farnsworth. Millions of Americans were formally introduced to the television at the 1939 World's Fair held in New York City, which had as its theme "The World of Tomorrow." Since an early retail television set cost about as much as an automobile, televisions remained a rarity in American households until the 1950s, when prices fell dramatically.

Television's ability to send a message around the world was demonstrated in July 1969. Hundreds of millions of people watched on live TV as astronaut Neil Armstrong stepped from the lunar module onto the surface of the Moon (Figure 1.14).

Television has created many opportunities for "news junkies" to get their fixes. The major commercial broadcast television networks have been supplemented by Fox, CNN, and other cable news organizations plus a myriad of Web sites. The various organizations compete with each other to be the first to break news stories. Increasingly, the media have turned to computer technology to help them provide information to the public. Sometimes this has led to embarrassing mistakes, as in the 2000 US presidential election.



FIGURE 1.13 Orson Welles's radio adaption of *War of the Worlds* on the evening of October 30, 1938, convinced many Americans that Martians were attacking New Jersey. (© Bettmann/Corbis)

At about 7:50 P.M. on the evening of Tuesday, November 7, 2000, before the polls had even closed in the Florida panhandle, the major networks began announcing that Al Gore would be the winner in Florida. Based on the expected result of the Florida election, the networks went on to predict—while people were still voting in the Western states—that Al Gore would be the next president of the United States.

You might be wondering how it is possible to predict the outcome of an election before everyone has voted. In a practice known as exit polling, a company called Voter News Service questions people leaving polling places. It combines the information it collects with early returns to predict the outcome of elections. Since 1988 the television networks have relied upon the Voter News Service to provide them with exit polling results.